

A DATA AND PROTECTION



1 Identity-disjoint data

MOBIO / LFW / FEI / SCface
Train / val / test people differ
Gallery never in exposures

2 Fixed face encoder

YuNet; 5-landmark alignment
ArcFace w600k_r50; 512-D
Unit main / raw separate study

3 Keyed protection

Template $T = f(\text{key}, x)$
Fresh / pool / shared key
Secret key values withheld

4 Exposure records

Input: batch $x \times n \times d$
8 nested sets per identity
 $n = 1, 2, 5, 10$ (by study)

OUTPUT d BioHash: 128 bits | MLP-Hash: 512 bits | IoM-GRP: 300 codes ($q=16$; one-hot $d=4,800$)
PolyProtect: 170 reals (window $m=5$, overlap=2). Scheme coverage differs by dataset.

B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



5 Supervised attacker

Single / mean / max MLP: $d \rightarrow 256 \rightarrow 512$
DeepSets: encode, mean, decode
Source embeddings: train targets only



6 Reconstructed embedding

L2-normalized prediction in 512-D
Train on non-test identities
Select checkpoint by validation loss



7 Held-out gallery linkage

Cosine scores vs. unprotected gallery
Top-1 / top-5 / AUROC / EER / TAR
Identity-clustered bootstrap intervals

KEY DESIGN Fresh: source-record keys are split-disjoint. Pool K: hidden transforms recur across identity splits.

REPLICATION MOBIO / FEI original; MOBIO / SCface extended: 2 splits x 3 seeds; 120-epoch caps.

DESIGN Three-pool prediction-mean baseline; raw learned and stricter native studies are separate controls.

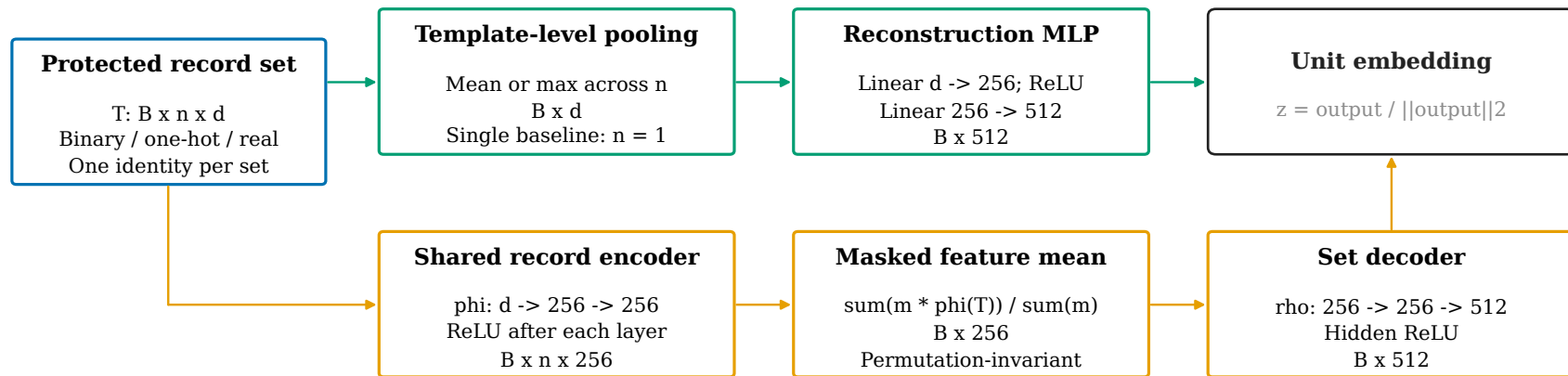
A MATCHED EXPOSURE CONSTRUCTION

One identity, n source records; nested subsets from one fixed permutation per repeat; gallery excluded.

Fresh: independent source-record keys. Pool K: hidden transforms recur across identity splits. No keys or slots in model input.

T: $B \times n \times d$ $d = 128$ BioHash; 512 MLP-Hash; 4,800 one-hot IoM-GRP; 170 PolyProtect

B TWO AGGREGATION PATHS



Primary: mean-pool MLP. Secondary: DeepSets. m masks valid records; all n records are valid in these runs.

C SUPERVISION

Target $u = \text{normalize}(\text{mean of exposed source embeddings})$.

Loss = $\text{mean}[1 - \cos(\text{angle}(z, u))] + 0.1 \times \text{MSE}(z, u)$.

Adam: lr 0.001; weight decay 0.0001; hidden width 256.

Train identities only; select minimum validation loss; freeze test.

D HELD-OUT IDENTITY LINKAGE

Scores = $z G^T$; G: $N \times 512$ unprotected unit gallery.

Rank identities; top-1/top-5, AUROC, EER, TAR at FAR.

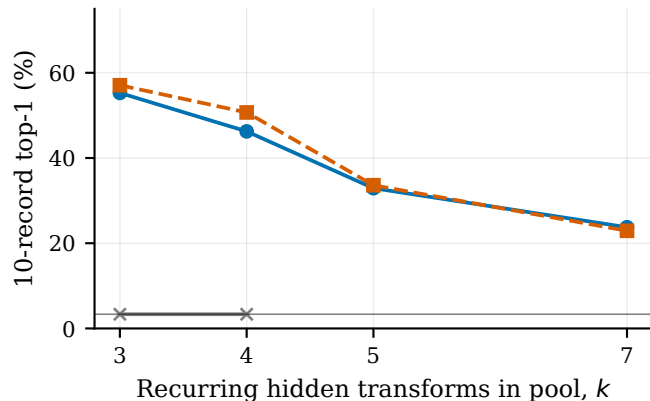
Follow-up: 3 model seeds x 2 identity partitions; 120 epochs.

Crossed seed/identity intervals; paired primary contrasts.

B: batch size; n: exposure count; d: template input width; N: test gallery identities. Source targets are not inference inputs.

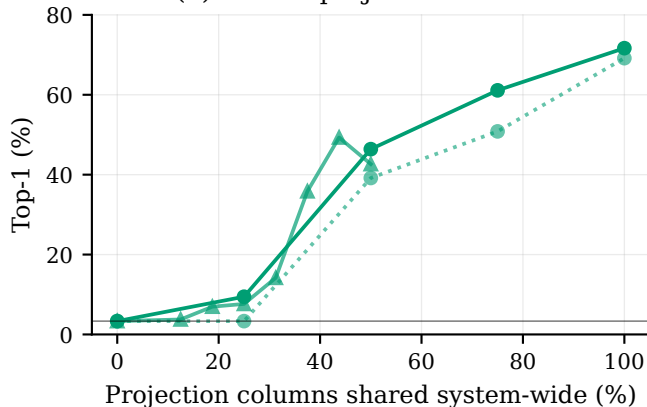
MOBIO mechanism controls (30 gallery identities; chance 3.33%)

(a) Slot label vs. shuffled records



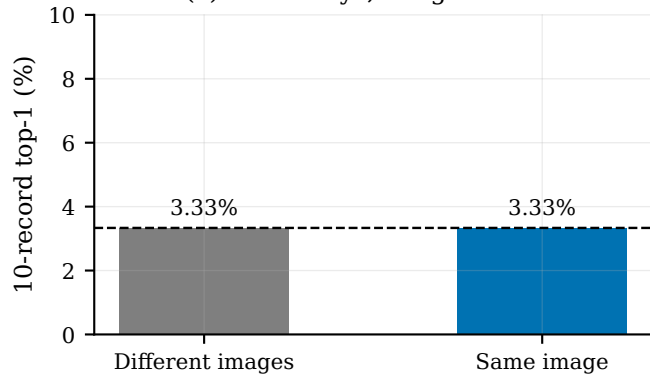
- slot hidden (DeepSets)
- slot label given to attacker
- records 2-10 from other identities

(b) Shared projection columns



- 10 records, coarse sweep
- 10 records, fine sweep (new split)
- 1 record, coarse

(c) Fresh keys, image control

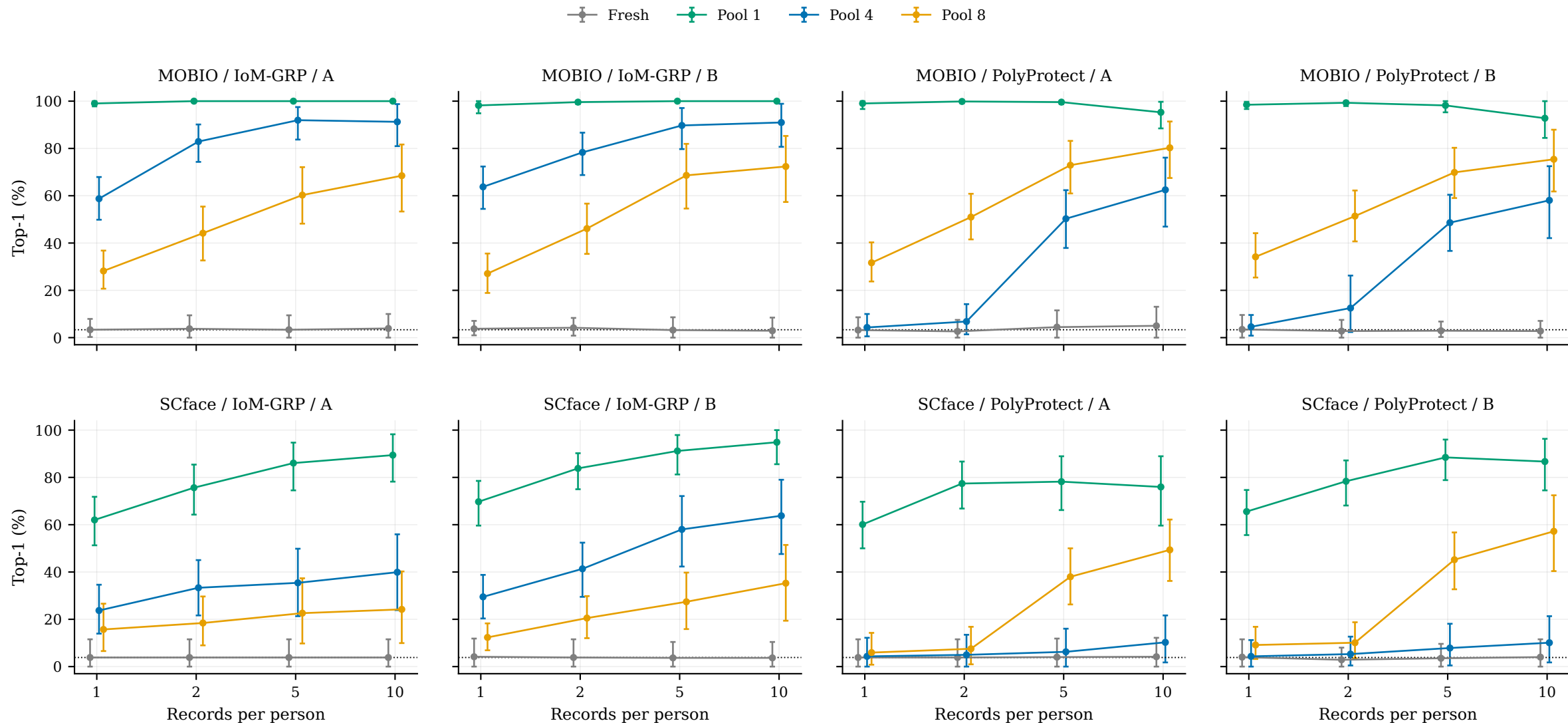


Dataset coverage: evidence layers are not interchangeable

BioHash historical	3 model seeds	3 model seeds	3 model seeds	3 model seeds
MLP-Hash historical	3 model seeds	Not run	Not run	Not run
IoM / PolyProtect pilots	1 model seed	Not run	1 model seed	1 model seed
Original scheme follow-up	3 seeds / 2 splits	Not run	3 seeds / 2 splits	Not run
Native / raw-norm audit	3 keys / 2 splits	Not run	3 keys / 2 splits	Not run
Independent-pool + baseline	3 pools / 3 seeds 2 splits	Not run	3 pools / 3 seeds 2 splits	Not run
Extended exposure study	3 seeds / 2 splits	Not run	Not run	3 seeds / 2 splits
Learned raw-input study	3 seeds / 1 split	Not run	Not run	3 seeds / 1 split
Stricter-selection audit	3 keys / 1 split	Not run	Not run	3 keys / 1 split
	MOBIO	LFW	FEI	SCface

SCface now has an extended study; coverage alone does not establish identical gallery protocols.

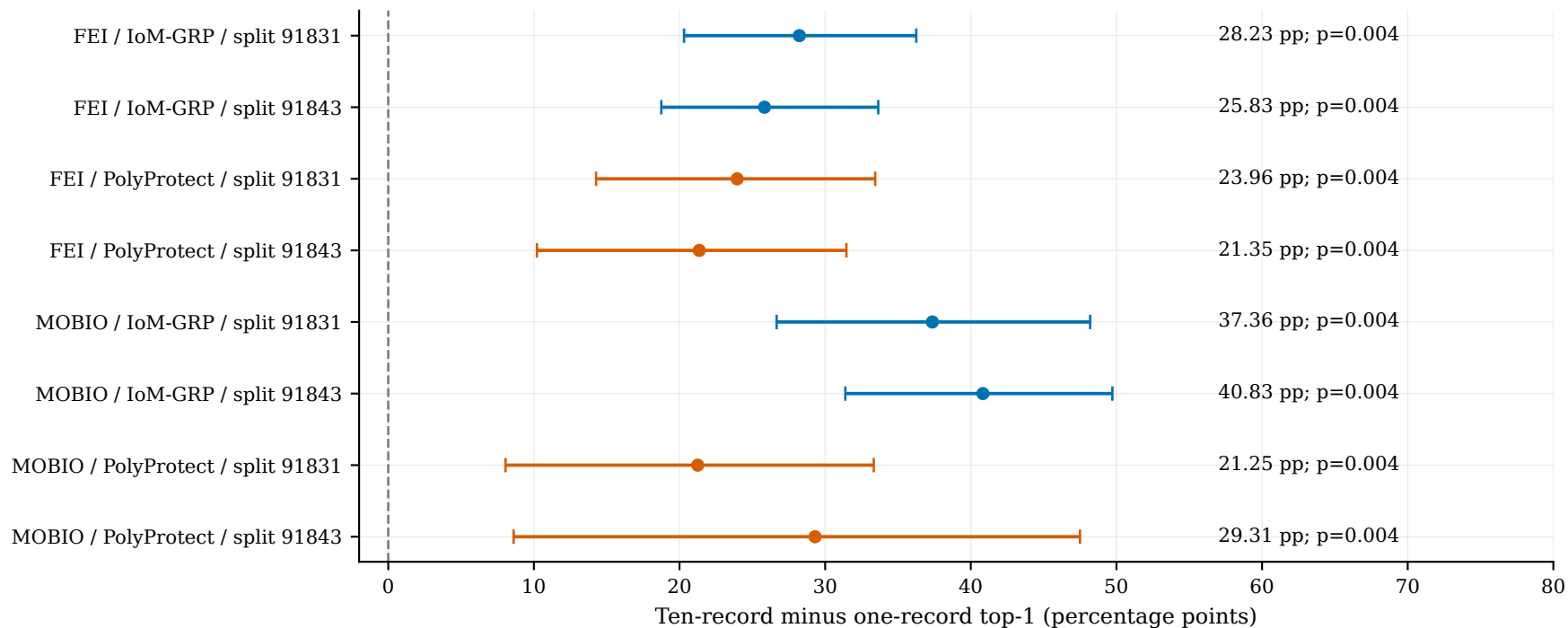
Grey = not run in this package, not zero accuracy. Identity partitions overlap; native audits do not train attackers.



Single MLP at 1 record; mean MLP at 2/5/10. Bars: crossed seed/identity 95% intervals; dotted line: chance.

A/B are overlapping identity assignments. One pool realization per condition; pool size is not an isolated causal effect.

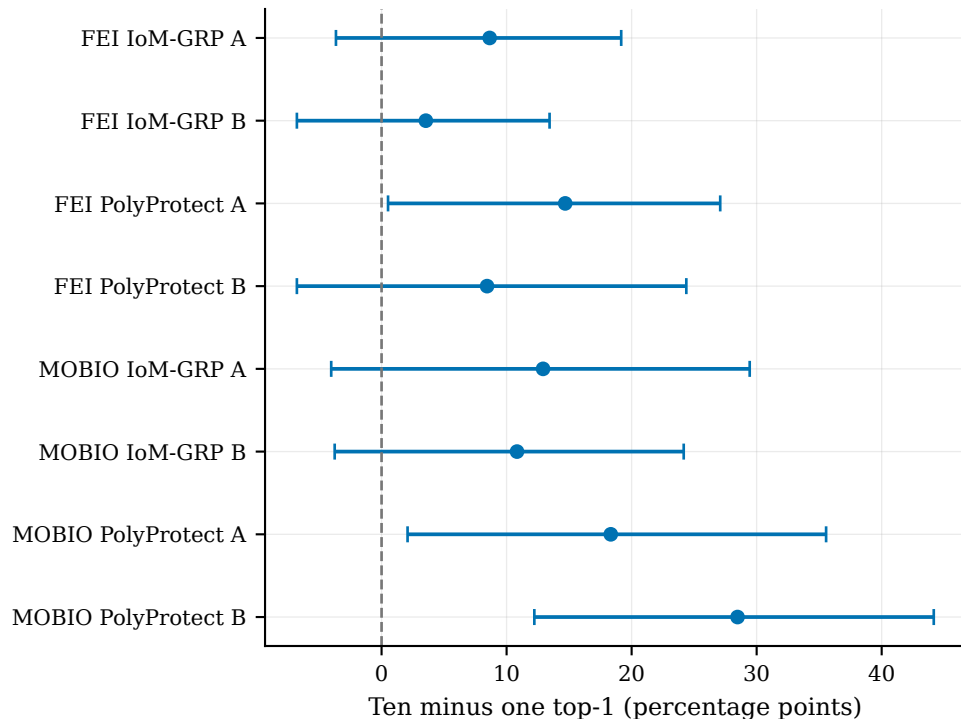
Pool-4 mean-pool amplification: three seeds, two identity partitions



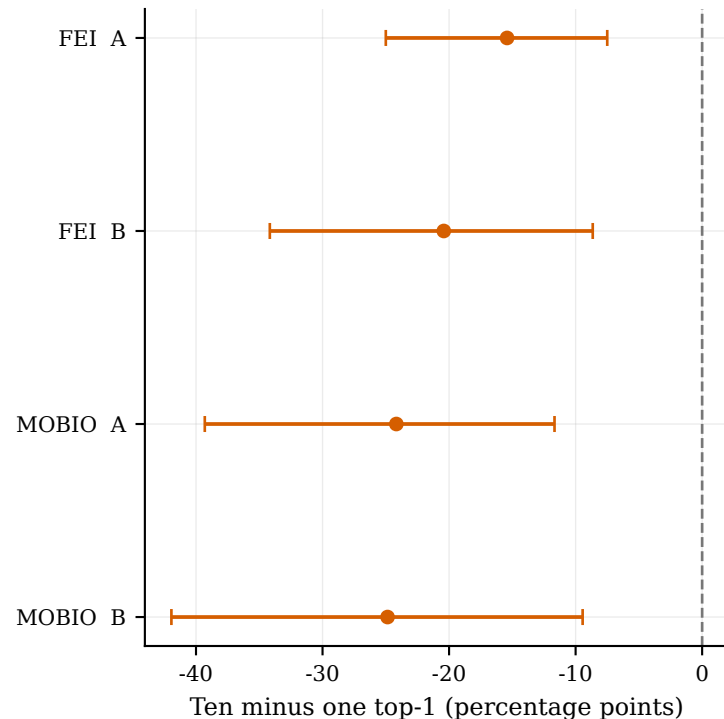
Bars: crossed model-seed / identity bootstrap 95% intervals. Tests: identity sign-flips, Holm family of 8.

Matched 120-epoch caps; key/set seeds fixed. Partitions are separate sensitivity analyses, not independent replications.

Pool-4 DeepSets: uncertain gains



Shared-key PolyProtect: regressions

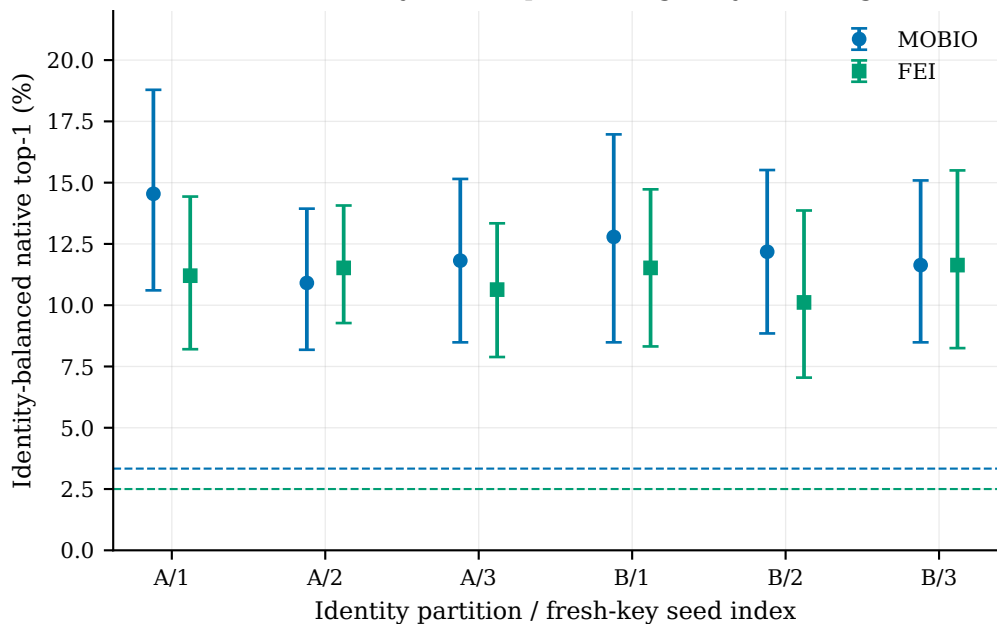


All 48 contrasts reported; post-hoc two-sided Holm family 48. Pointwise crossed seed/identity 95% intervals.

Left: none survives correction (minimum $p = 0.0576$). Right: all four negative contrasts survive ($p = 0.0192$).

All eight pool-4 mean-pool gains remain significant ($p = 0.0192$). More records do not universally improve linkage.

(a) Fresh PolyProtect: protected-gallery matching

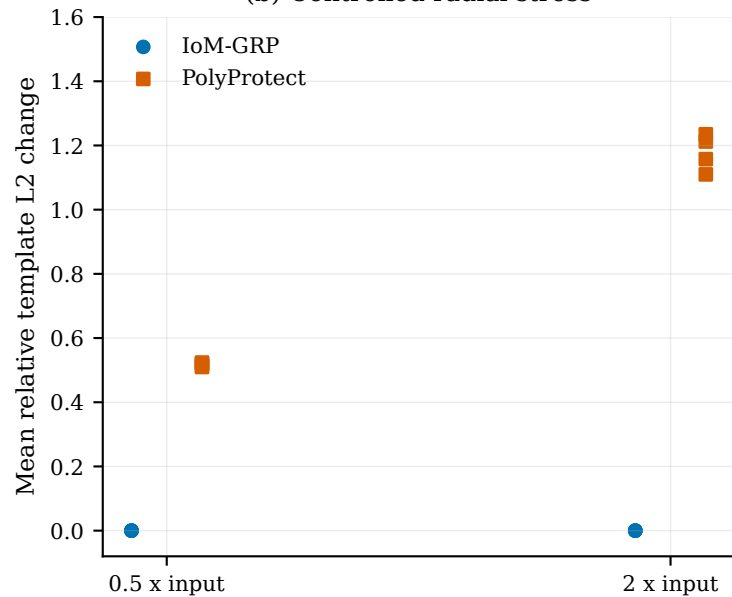


Native tests: 12 / 12 Holm-adjusted $p \leq 0.006$; identity-clustered 95% intervals; dashed lines: chance.

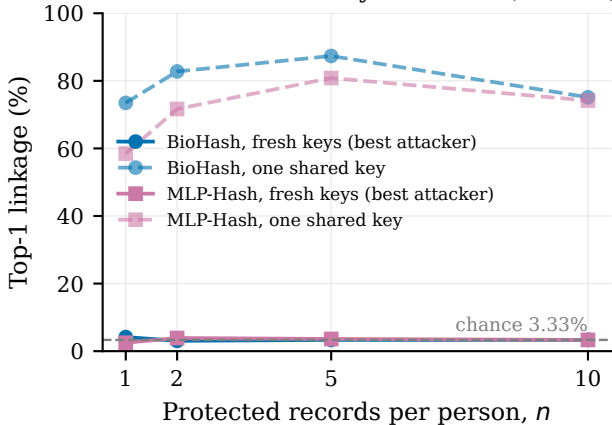
A/B: splits 91831/91843. Key seeds 1/2/3: 91873/91879/91883. Gallery-label permutations preserve probe clusters.

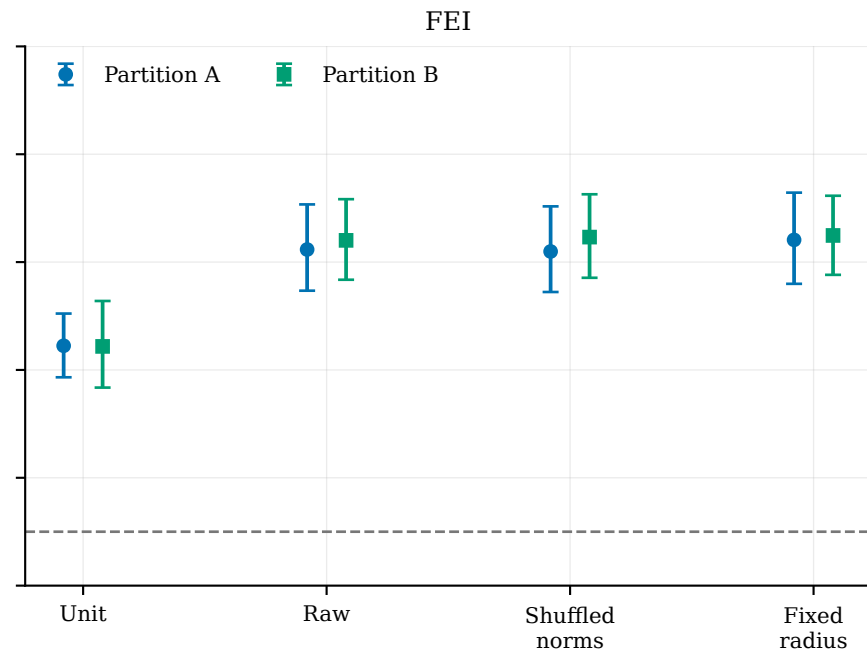
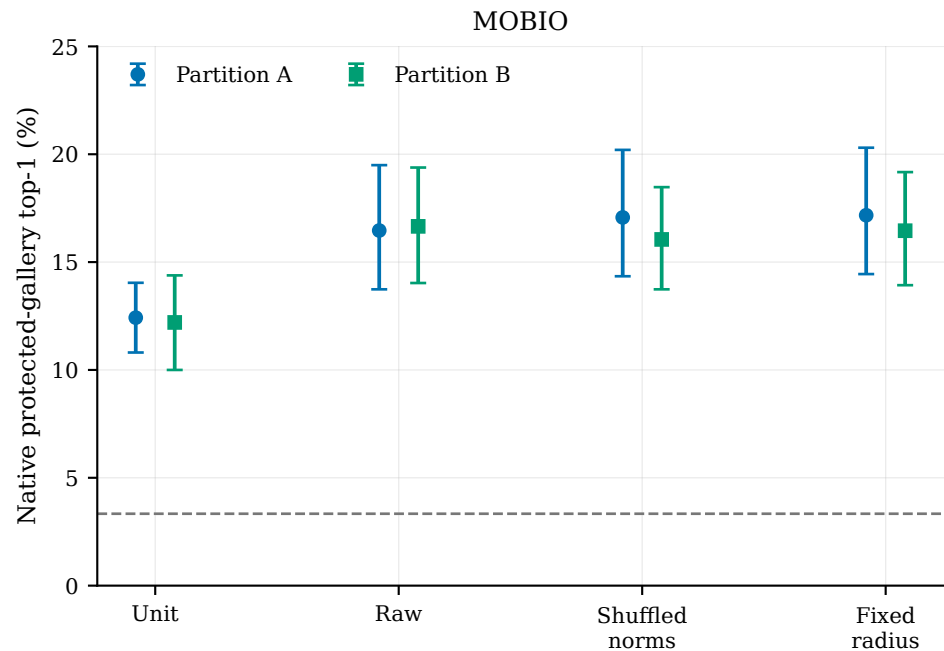
Radial stress: 32 held-out records per dataset/partition, fixed keys; measures scale sensitivity, not natural norm leakage.

(b) Controlled radial stress



Fresh and shared keys: MOBIO (N = 30)



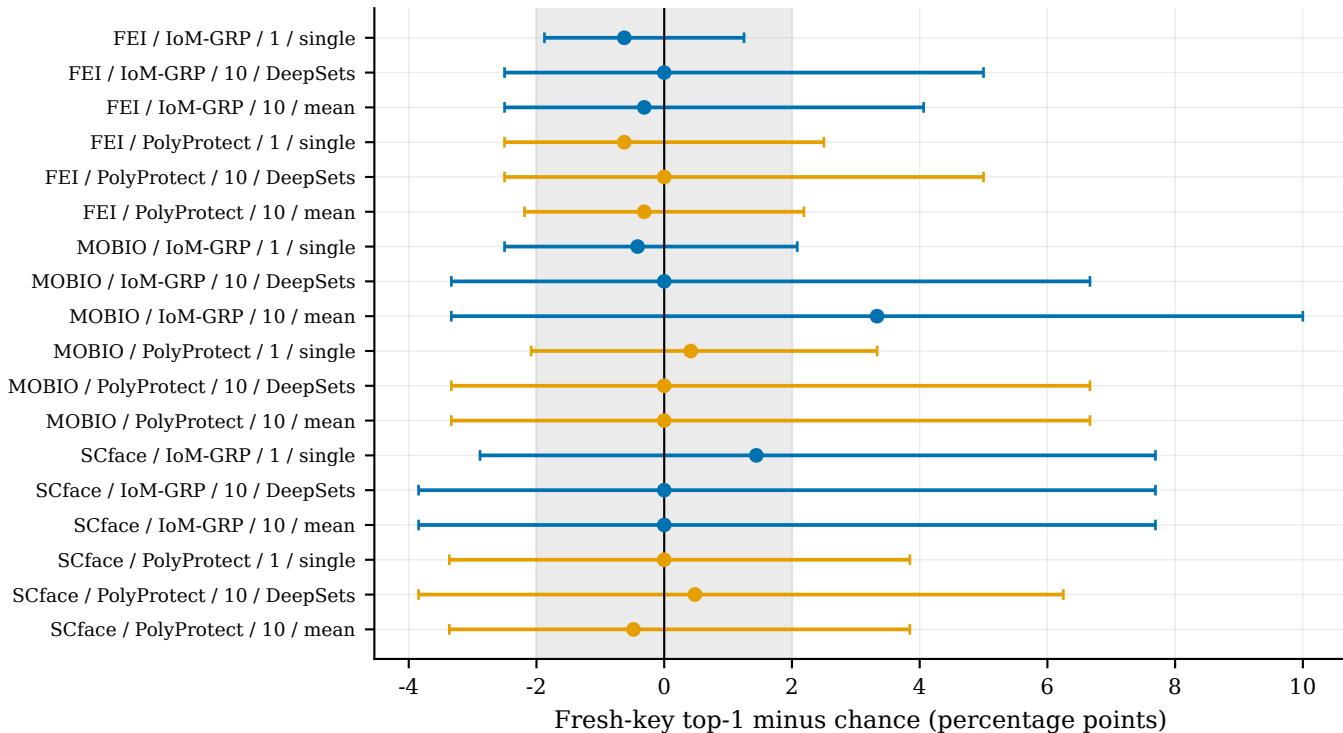


Three fixed key seeds averaged; identity-bootstrap 95% intervals. All 16 null tests: Holm $p = 0.0032$.

Raw minus unit survives paired correction only on FEI. Raw minus shuffled: all adjusted $p \geq 0.7584$.

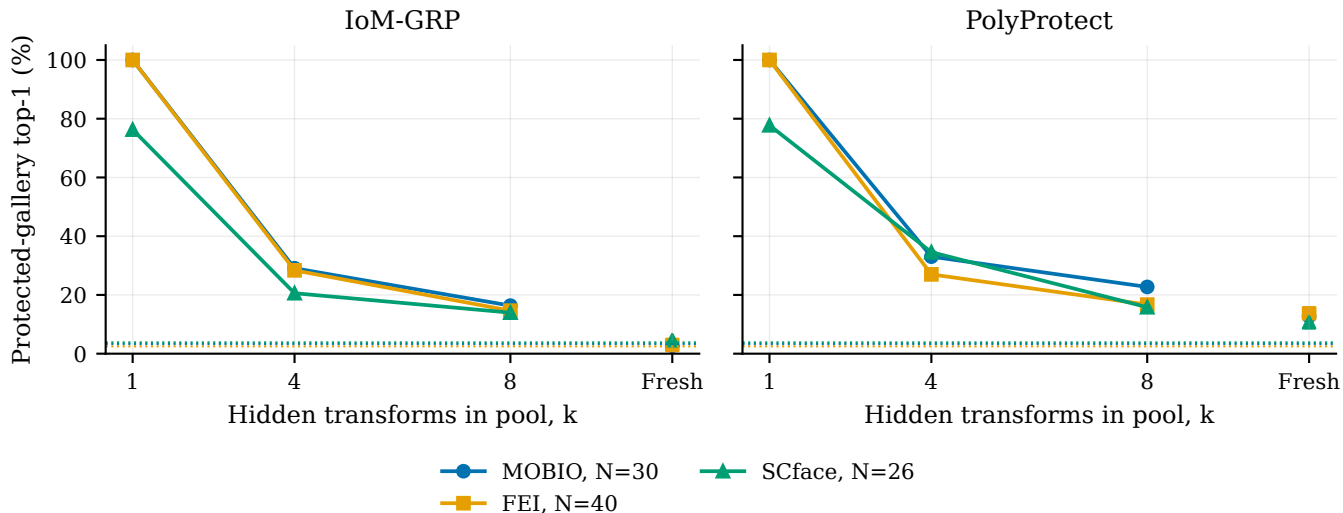
Natural identity-specific norm leakage is not established. No learned raw-input attack was retrained.

Fresh-key 90% intervals: exploratory pilot sensitivity

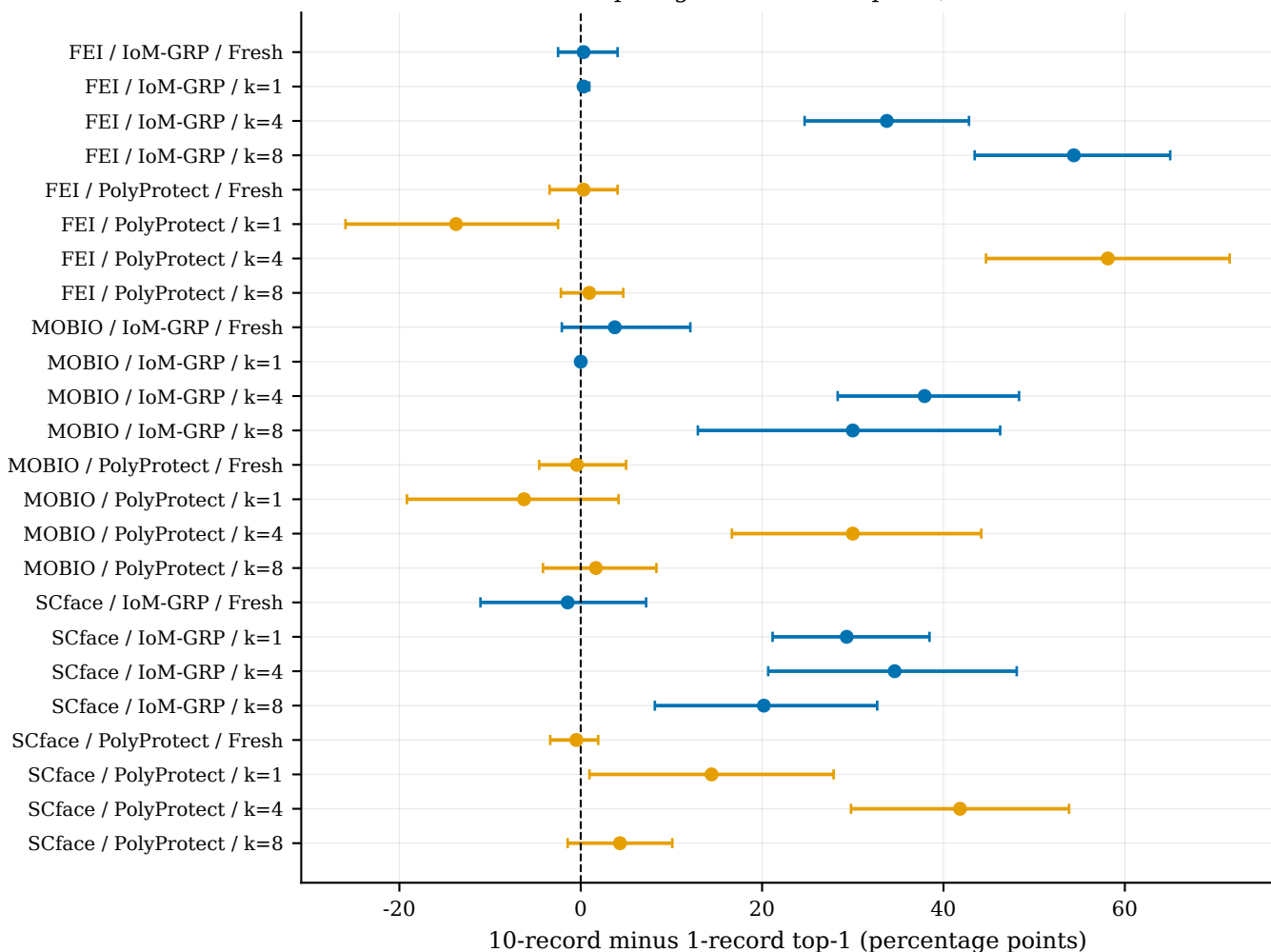


Illustrative +/-2 pp band; not an approved equivalence margin

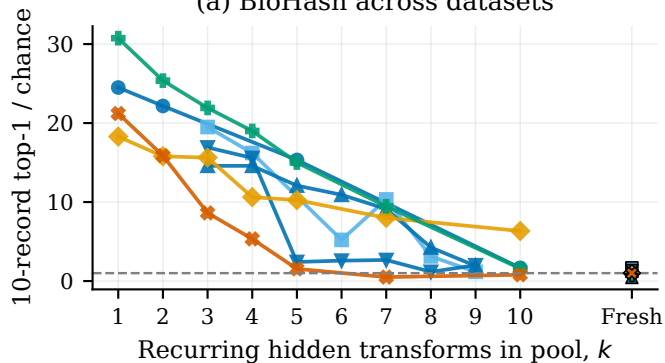
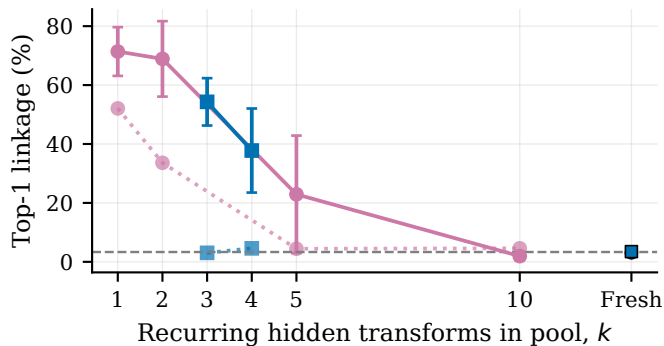
Native matching diagnostic, not learned embedding linkage



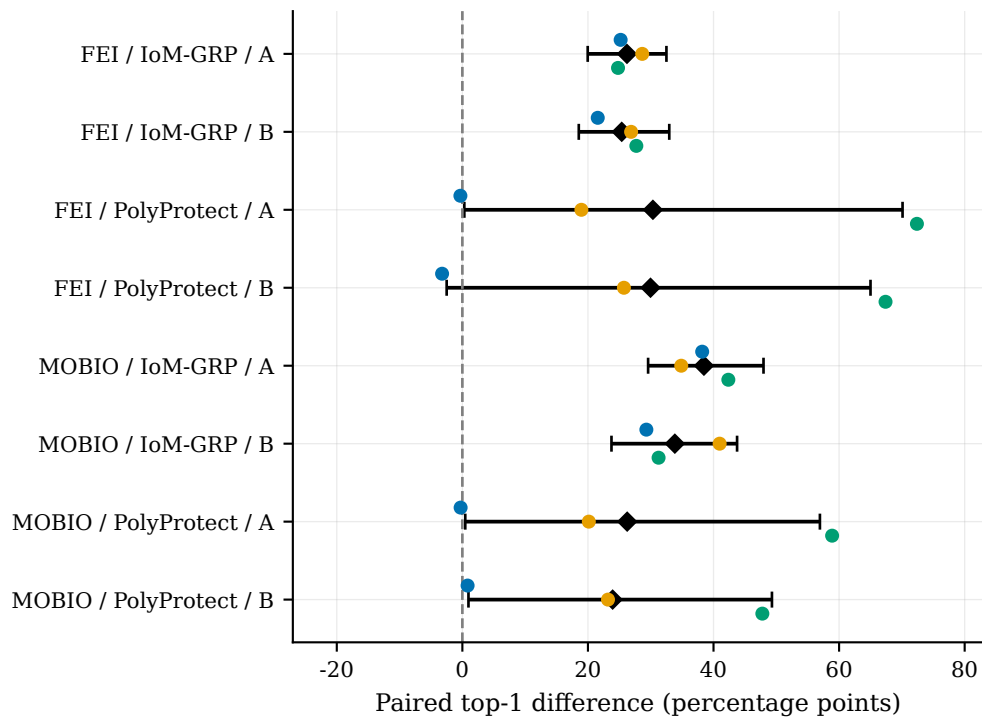
Paired mean-pool gains: one-seed pilots, 95% intervals



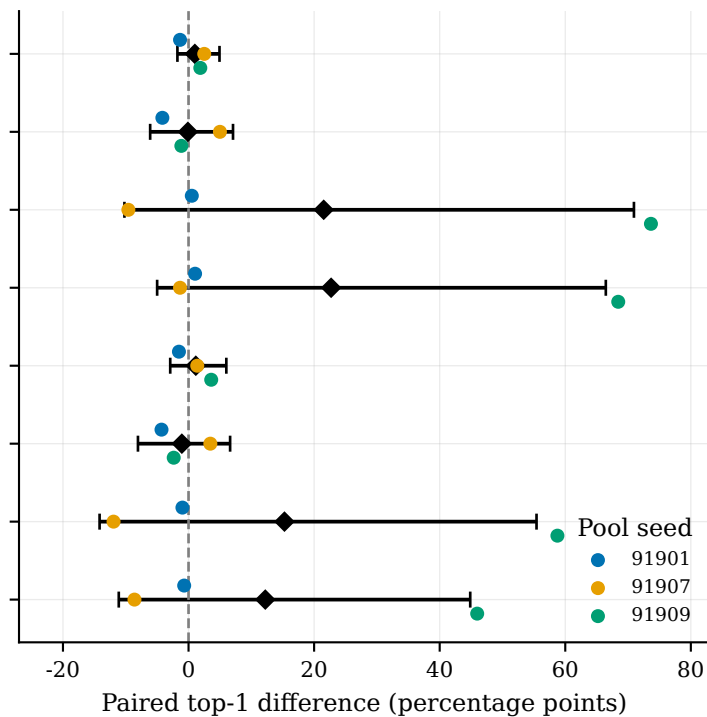
(a) BioHash across datasets

(b) MLP-Hash, MOBIO ($N = 30$)

(a) Ten-record gain over one record



(b) Input mean minus prediction mean

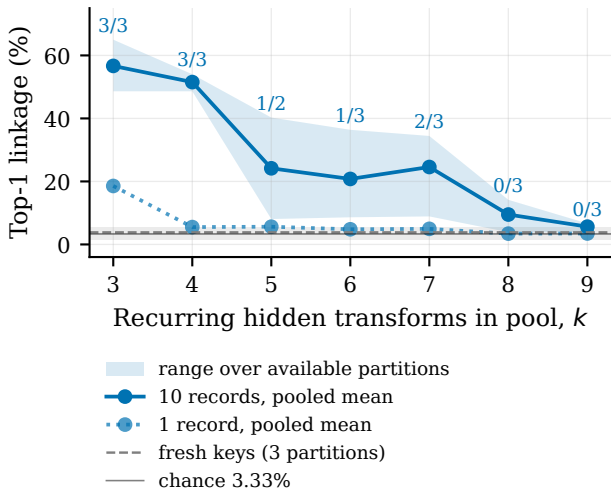


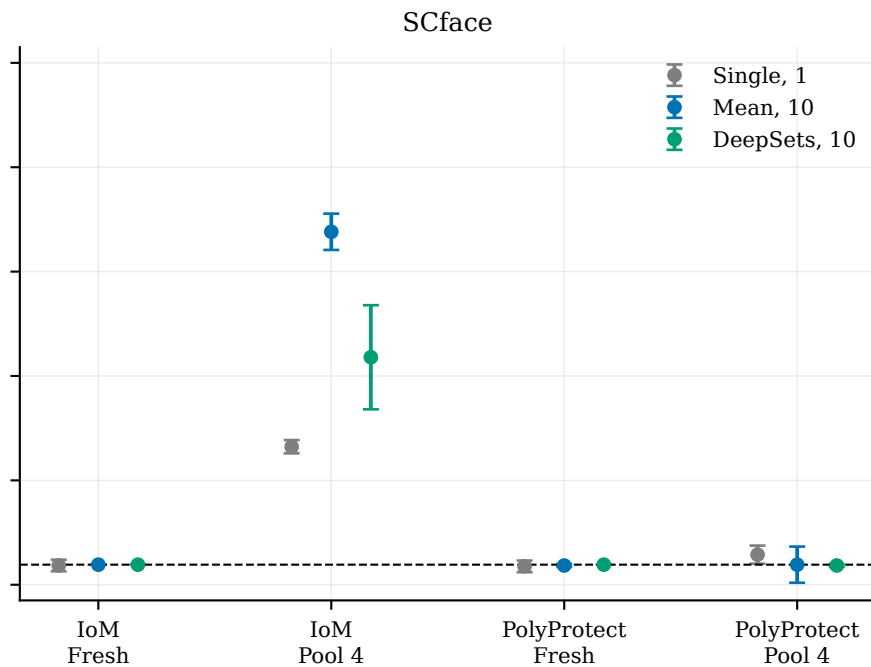
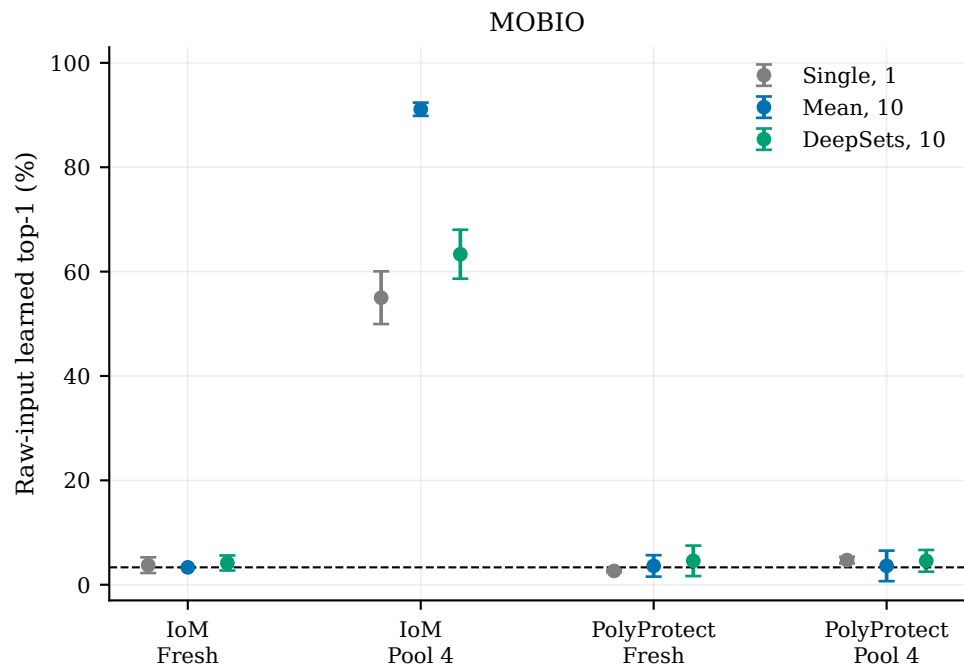
Colored dots: individual pool draws, averaged over three model seeds. Black: mean and crossed pool/seed/identity 95% interval.

IoM gains persist across draws; PolyProtect is pool-sensitive. Every direct baseline-comparison interval includes zero.

Three pools only; pointwise uncertainty, no corrected significance claim. A/B: overlapping partitions 91831/91843.

MOBIO boundary (N = 30)





Bars: standard deviation across three model seeds, NOT confidence intervals. One saved identity split; descriptive study.

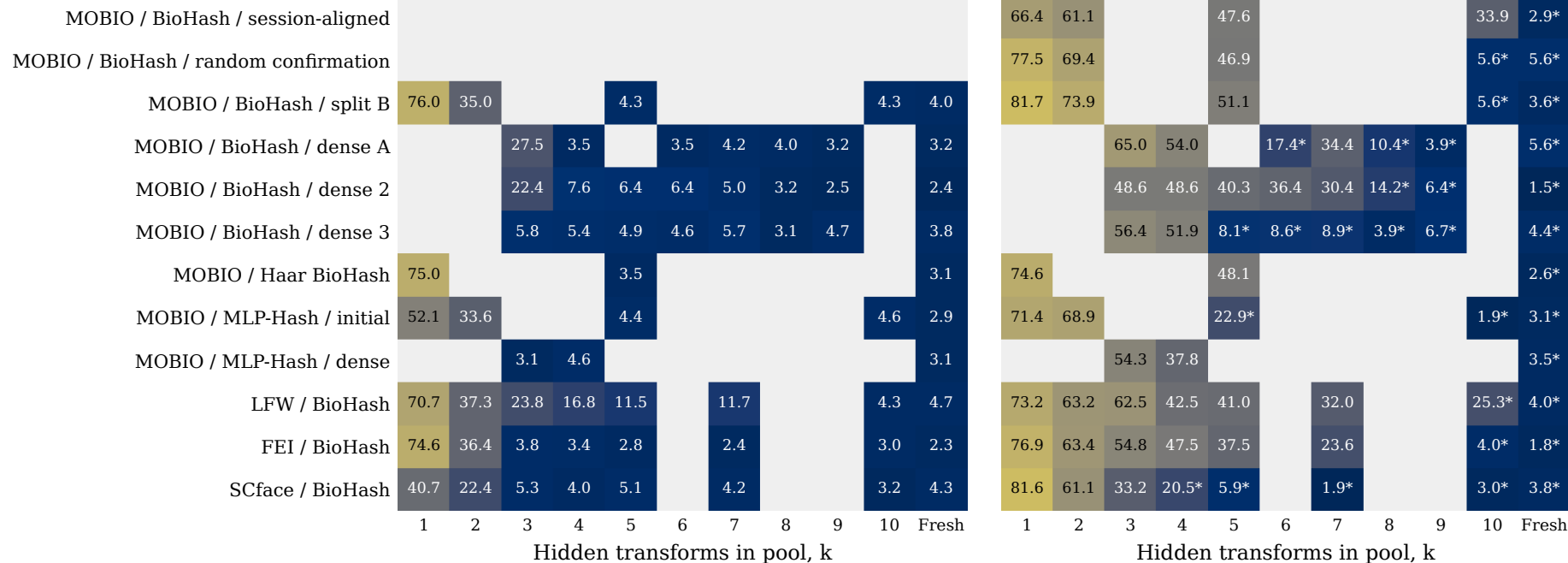
PolyProtect is near chance in this run; IoM retains large pool-4 gains. No matched unit-input retraining at these seeds.

Different pool/set seeds and partitions prevent attributing cross-study accuracy differences solely to normalization.

Earlier three-seed key-pool studies

(a) One record

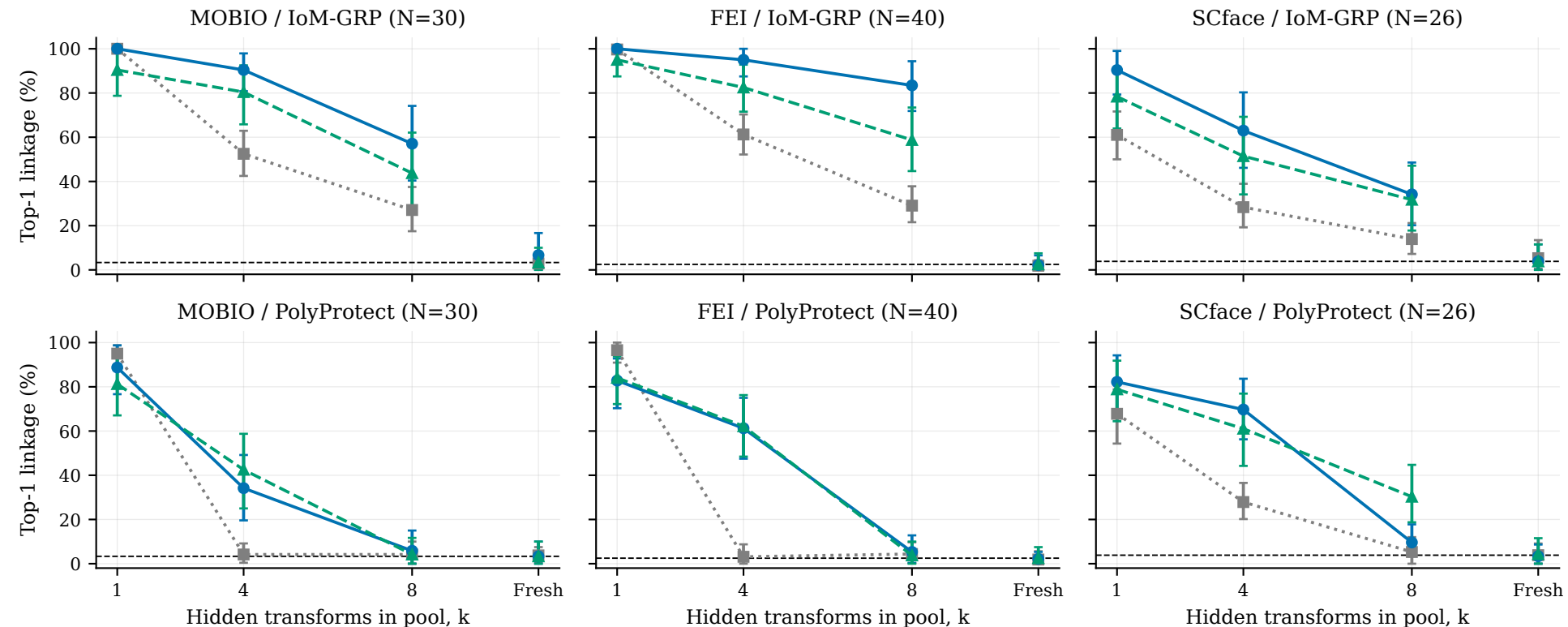
(b) Ten records



Cells: top-1 (%), 3-seed means. Grey: unavailable. *: not all seed intervals exclude chance.
 Gallery sizes: MOBIO 30; LFW 25; FEI 40; SCface 26. Rows are separate studies, not matched replications.

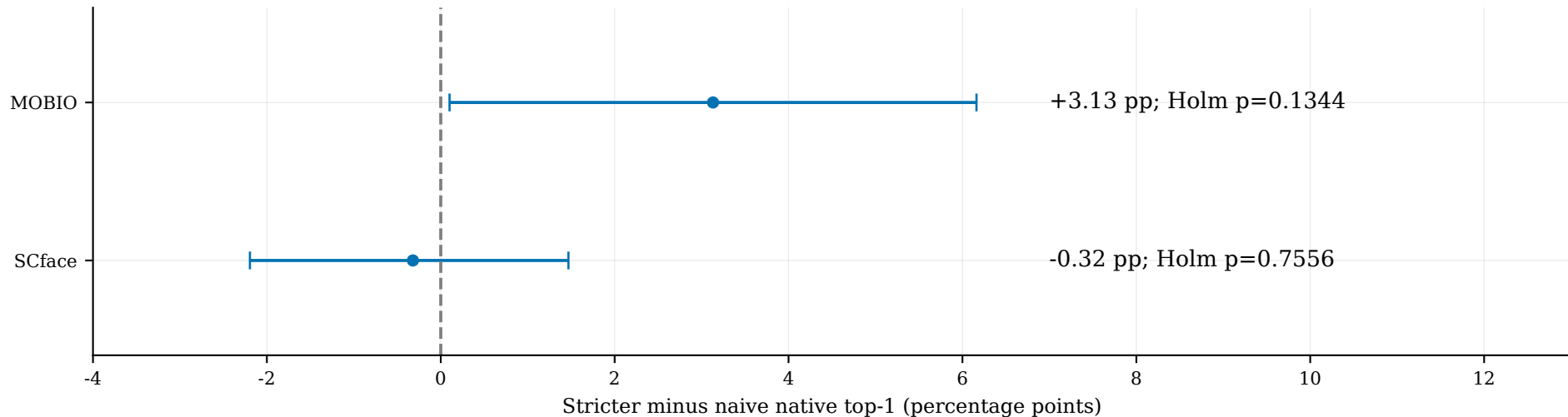
New protection schemes: one-seed pilots

120-epoch cap; bars: identity-clustered 95% intervals; dashed baseline: chance



1 record 10 records, mean 10 records, DeepSets

Local stricter selection: no corrected evidence of reduced linkage



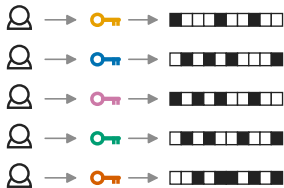
Left of zero would mean less native linkage. Bars: identity-bootstrap 95% intervals, conditional on three key seeds.

20 candidates; 200 development pairs; score band $[-0.5, 0.5]$. Paired Holm family 2; neither comparison passes 0.05.

Local operationalization, not source-exact replication or a refutation of all stricter PolyProtect parameter policies.

Fresh keys

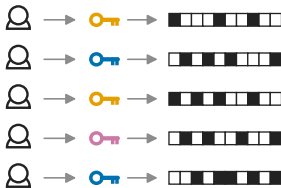
one key per source record



no transform reuse

Recurring key pool

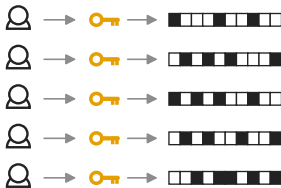
k transforms; slot labels hidden



same colour = same transform
reuse across train and test

Single shared key

k = 1; key value still hidden



shared-transform baseline

Learned attacks inconclusive

PolyProtect outside the theorem

Multi-record gains observed

boundary depends on dataset and split

Strong linkage observed

hidden does not mean unlearnable

Protocol

Disjoint train / validation / test identities; one held-out gallery image per identity
8 sets per identity; 3 seeds earlier, 1 in pilots; identity-clustered intervals